

Total No. of Questions : 8]

SEAT No. :

PE-2234

[Total No. of Pages : 2

[6584]-135

B.E. (Electronics and Telecommunication Engineering)
VLSI DESIGN AND TECHNOLOGY
(2019 Pattern) (Semester - VII) (404182)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Assume suitable data, if necessary.

Q1) a) Write the features of CPLD in detail and give it's applications. [8]

b) Draw and explain the architecture of FPGA & compare between CPLD and FPGA. [10]

OR

Q2) a) Draw and explain macrocell in detail. [8]

b) Draw and explain PLD design flow. [10]

Q3) a) Draw and explain CMOS Inverter schematic and it's voltage transfer characteristics. [9]

b) What are the merits of Transmission gate? Design 2:1 multiplexer using transmission gate. [8]

OR

Q4) a) Write short note on channel length modulation. [6]

b) Write short note on Device sizer. [6]

c) Write short note on velocity saturation. [5]

Q5) a) What is DRC? Explain in detail rules in CMOS VLSI design? [9]

b) Draw stick diagram for Inverter, NAND and NOR gate. [9]

OR

Q6) a) Write short note on Design issues like Antenna Effect. [6]

b) Write short note on SPICE simulation. [6]

c) Write short note on Layout Vs schematic. [6]

P.T.O.

- Q7)** a) Explain the need for Design for Testability? Explain stuck-at-0 and stuck-at-1 faults with example. [9]
- b) Write short note on [8]
- i) JTAG
- ii) Boundary scan

OR

- Q8)** a) What is the need of BIST? Explain typical BIST in detail. [9]
- b) Draw the TAP controller state diagram and explain. [8]

